Modelling Software-based Systems:
Correct by Construction Paradigm
Chapter *The Modelling Language Event-B*
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Design of Correct by Construction Sequential Algorithms

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1.1. Introduction

The development of correct algorithms or sequential programs from specifications (Dijkstra 1976) is a scientific theme linked to that of the verification of programs or algorithms (Turing 1949 ; Floyd 1967 ; Hoare 1969). The fundamental question can be summarised in the form of a symbolic equation $\mathsf{D}, \mathsf{A} \Rightarrow \mathsf{C}$ where D (resp. A, C) is the problem domain (resp. the algorithm, the contract). In this equation, we assume that the problem domain D is known and may be, for example, $\mathbb{Z}$, the domain of integers, and we will be interested in problems requiring properties on integers. A problem is a general expression to designate the calculation of a value from data or the search for a value in a set of data. The algorithm A is expressing assignment instructions, conditional instructions and bounded or unbounded iterations. Finally, the contract C is defined by two elements a pre-condition $\mathsf{pre}(v_0)$ and a post-condition $\mathsf{post}(v_0, v_f)$ relating the initial value v_0 of a variable v to its final value v_f. Solving the problem consists in expressing it by a contract ensuring that for any initial value v_0 satisfying $\mathsf{pre}(v_0)$, there exists a value v_f satisfying $\mathsf{post}(v_0, v_f)$. On the other hand, it is important that the final value of v_f corresponds to a calculation of an algorithm A in the classical sense of computability (Rogers 1967). The equation can therefore be

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coordinated by The Modelling Language Event-B .

rewritten in the following form:$\forall v_0, v_f \in \mathsf{D}.\mathsf{pre}(v_0) \wedge v_0 \xrightarrow{\mathrm{A}} v_f \Rightarrow \mathsf{post}(v_0, v_f)$ and we obtain the expression for the partial correctness of the algorithm A in relation to the contract $\mathsf{C}(v, \mathsf{pre}(v_0), \mathsf{post}(v_0, v_f))$ on the domain D. The relation $\xrightarrow{\mathrm{A}}$ expresses the calculation of A and we can add a second expression which plays the role of the termination of A: $\forall v_0 \in \mathsf{D}.\mathsf{pre}(v_0) \Rightarrow \exists v_f \in \mathsf{D}.v_0 \xrightarrow{\mathrm{A}} v_f$. The two translations produce a synthetic expression of the following form:

$$\forall v_0 \in \mathsf{D}.\mathsf{pre}(v_0) \Rightarrow \begin{pmatrix} \forall v_f \in \mathsf{D}.v_0 \xrightarrow{\mathrm{A}} v_f \Rightarrow \mathsf{post}(v_0, v_f) \\ \exists v_f \in \mathsf{D}.v_0 \xrightarrow{\mathrm{A}} v_f \end{pmatrix}$$

which we rewrite with the wp calculus as follows: $\forall v_0 \in \mathsf{D}.v = v_0 \wedge \mathsf{pre}(v_0) \Rightarrow wp(A)(\mathsf{post}(v0, v))$

This discussion led us to give meaning to the correctness of an algorithm by considering its partial correctness as well as its termination. Hoare logic most often expresses partial correctness and in all rigour it would be necessary to use two notations, one expressing partial correctness and the other total correctness, but the objective here is not to verify an algorithm A and therefore to verify a list of verification conditions as Floyd method indicates, but to find an algorithm A which satisfies this equation $\forall v_0 \in \mathsf{D}.v = v_0 \wedge \mathsf{pre}(v_0) \Rightarrow wp(A)(\mathsf{post}(v0, v))$. In the *a posteriori* approach to correcting algorithms, we propose a solution for A and then check a list of verification conditions. This technique also consists of checking the verification conditions without having clearly stated the contract C.

Semantic analysis techniques can thus be developed with the abstract interpretation (Cousot 2021) and this is based on semantic techniques that simplify the life of the programmer who obtains analysis feedback. Correctness by construction is a technique which starts with an abstract algorithm A_0 which fulfils the contract in a verified way and which is progressively enriched with increasingly complex control structures, while preserving the property of correctness with respect to the contract C. This progressive strategy of adding elements to make the result more precise is guided by the refinement between the algorithms. Each transformation or refinement step guarantees that the resulting algorithm is correct. C. Morgan (Morgan 1990) develops the refinement calculus which makes it possible to progressively and correctly transform one algorithm into another algorithm. The operation guarantees that the final algorithm is correct with respect to the first algorithm which is a pre/post specification considered as an algorithmic action of the form $v : |(pre, post)$. $v : |(pre, post)$ designates an algorithmic statement I whose effect is to modify v by respecting the contract defined by pre and post. Thus, the strategy consists of constructing a sequence of algorithms A_0,..., A_i, ..., A_n with the properties:

– A_0 is the expression of the contract: $\mathsf{D}, \mathsf{A}_0 \Rightarrow \mathsf{C}$

– for all i in $1..n$, the algorithm A_i refines A_{i-1}: $\mathsf{D}, \mathsf{A}_i \Rightarrow \mathsf{C}$ and $\mathsf{D}, \mathsf{A}_{i-1} \Rightarrow \mathsf{C}$.

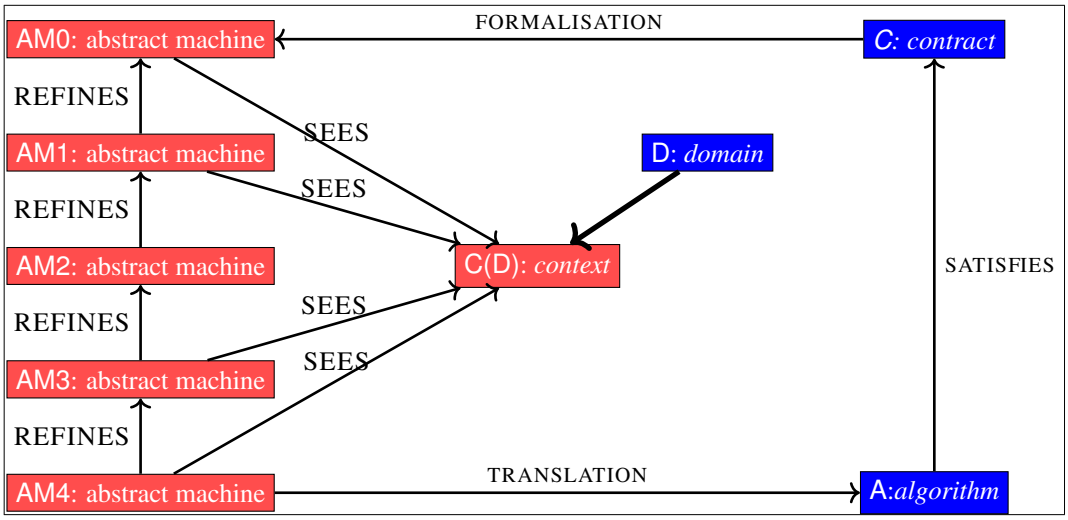

Figure 1.1. *Correctness by construction in Event-B*

More recently, Derrick G. Kourie and Bruce W. Watson (Kourie and Watson 2012) follow this strategy and implement the correction-by-construction paradigm on classical programming problems. We have identified in these two calculations of the fact that the contract becomes an algorithmic statement corresponding to a generalised algorithmic structure. In fact, as we can see, a contract can express the halting problem of Turing machines (Rogers 1967) in a language of assertions which is still abstract, but this does not mean that we have solved the halting problem, but that we have extended the algorithmic language with a statement *magic* enabling it to be solved. This amounts to extending the space of solutions and then choosing what corresponds to the theory of computability. C. Morgan reminded us that all second degree equations have solutions in fields of complexes, but that the method of solving in the set of reals retains only the real solutions. The specification statement $v : [\mathsf{pre}, \mathsf{post}]$ is a valid statement in this algorithmic language. Such a specification instruction can be expressed in the Event-B language.

This approach to developing correct by construction algorithms is quite simply implemented in the Event-B language and in fact equipped by the Rodin environment. Figure 1.1 describes the general idea. This idea consists of translating the contract as an *event* which performs the calculation described by the contract. The contract expresses the *what* but carries out the computation as an observation of the event. This event is placed in a first abstract machine AM0, which uses the mathematical elements extracted from the problem domain D and expressed in the context Event-B

C(D). The problem domain D is a structure with data and properties related to the problem to solve and it may be generally naturals or integers for arithmetic, arrays for searching or sorting... C(D) is a context designed from the problem domain by adding axioms and theorems for sets and constants that are necessary for the cdrivation of Event-B machines. The development of an algorithm consists in the gradual enrichment of the extracted machines AM0, ... AMn, by expressing the computations necessary to systematically translate the last machine into an algorithmic form so as to guarantee the correctness of the algorithm A, thanks to the correctness of the transformations provided by the refinement. It should be noted that correctness concerns partial correctness and termination, and that the abstract machines are models containing variables that will not be necessarly present in the algorithm produced.

We will present two techniques that implement this development method:

– the inductive pattern based on transformations of abstract machines by Jean-Raymond Abrial (Abrial 2010, Chapter 15).

– the recursive pattern based on the relation *call* as *event* that we have developed (Méry 2009 ; Cheng *et al*. 2016).

We will present these two methods by highlighting case studies of classical sequential algorithms.

1.2. Design of a Iterative Sequential Algorithm

1.2.1. *Problem 1 : Calculating the sum of a vector v of integer values*

First, we define the contract associated with this problem of calculating the sum of the elements of the vector v_0. The algorithm we are looking for is SUM.

The domain of the problem to be solved is that of sequences of integers $\mathbb{SEQ}(\mathbb{Z})$ and the contract states that the value of the result is the sum of the integers in the sequence v. This mathematical expression is not directly expressible in the mathematical language of Event-B and we define a sequence u characterising the values of the partial sums. The context associated with our C($\mathbb{Z}$) Event-B model is defined by enumerating the *requires* hypotheses and defining u.

First, we need to express the sum r of the sequence v_0 in the language of Event-B ; this formulation is immediate in mathematical terms: $r = \sum_{k=1}^{k=n_0} v_0(k)$. As the notation for summing a finite sequence of values is not provided in the basic elements of the language, we must *define* this notion in a context *c0* which will contain the data of the problem and the notations defined specifically for this case.

variables n, v, r

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requires
$$\begin{cases} n_0 \in \mathbb{N} \land n_0 \neq 0 \\ v_0 \in 1..n_0 \rightarrow \mathbb{Z} \end{cases}$$

ensures
$$\begin{cases} r_f = \sum_{k=1}^{k=n_0} v_0(k) \\ n_f = n_0 \\ v_f = v_0 \end{cases}$$

Thus, the *data* n_0 and v_0 are defined as being respectively a non-zero natural integer (axioms pre1, pre2) and a function v_0 of domain $1..n_0$ and codomain $\mathbb{SEQ}(\mathbb{Z})$ (axiom pre3). The aim is to define the theory in which we will describe our data.
The naming of axioms with prefix *pre* allows to declare the three first axioms as given facts of the precondition. From a programming viewpoint, they define the contract signature, ie the types of the input parameters of the function to be implemented.

Secondly, we introduce a sequence u of integer values corresponding to the partial sums $\sum_{k=1}^{k=i} v_0(k)$. To do this, the idea is to define the partial sums using an inductive definition, which technically requires us to be sure of the *well definition* of this sequence u. The sequence u is therefore defined as follows:

– u is a total function from $0..n0$ to $\mathbb{Z}$ (axiom axm1).

– Initially, the summation starts with 0 and $u(0) = 0$ (axiom axm2).

– For values of k less than n_0, the value of $u(k)$ is defined from that of $u(k-1)$ and $v_0(k)$ (axiom axm3).

Axioms are given in the context *S0* and constitute a logical and mathematical theory, which will be useful for proving the properties of the models we will develop later.

CONTEXT $S0$
CONSTANTS $n0\ v0\ u$
AXIOMS
$@pre1\ n0 \in \mathbb{N}$
$@pre2\ n0 \neq 0$
$@pre3\ v0 \in 1..n0 \rightarrow \mathbb{Z}$
$@axm1\ u \in 0..n0 \rightarrow \mathbb{Z}$
$@axm2\ u(0) = 0$
$@axm3\ \forall k.\ k \in 1..n0 \Rightarrow u(k) = u(k-1) + v0(k)$
END

Proof obligations (WD) ensure the well-definedness of axioms. We have therefore defined the mathematical framework of the problem and we will now define the problem of summing the sequence v_0.

1.2.1.1. *Specification of the problem to solve*

The problem is therefore to calculate the value of the sum of the sequence v. We define a machine *S1* which is an abstract machine expressing through the event final, the expression of the *postcondition* $r = u(n)$. In fact, the new value of the variable r will be $u(n)$, when the event final will be observed. The initial value of r is arbitrary at initialisation.

MACHINE $S1$ SEES $S0$
VARIABLES $r\ v\ n$
INVARIANTS
$@inv1\ r \in \mathbb{Z}$
$@inv2\ n \in \mathbb{Z}$
$@inv3\ v \in 1..n \rightarrow \mathbb{Z}$
$@read - values\ n = n0 \ \wedge \ v = v0$
EVENTS
EVENT $INITIALISATION$
then
$@act1\ r :\in \mathbb{Z}$
$@act2\ n := n0$
$@act3\ v := v0$
end
EVENT $final$
then
$@act1\ r := u(n)$
end
$anticipated$ EVENT $keep$
then
$@act1\ r, n, v :|$
$(\ r' \in \mathbb{Z} \wedge n' \in \mathbb{Z}$
$\wedge\ v' \in 1..n' \rightarrow \mathbb{Z}$
$\wedge\ n' = n0 \ \wedge \ v' = v0\ )$
end
END

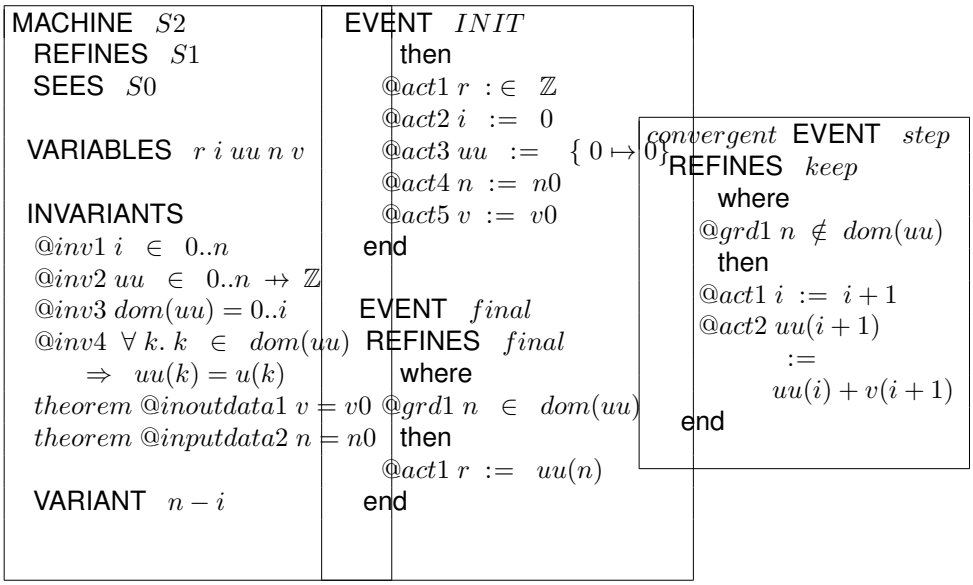
Finally, the variable r must satisfy the invariant $inv1 : r \in \mathbb{Z}$ which specifies the type of r. The event final is therefore simply an assignment of the value $u(n)$ to r.
We can express it as a HOARE triple: $\{n = n_0 \wedge v = v_0 \wedge n_0 > 0 \wedge v_0 \in 1\,..\,n_0 \rightarrow \mathbb{N}\}\mathsf{SUM}\{r = u(n_0)\}$.

Note that the data is *visible* (sees) from the context *S0*. The problem is therefore to find an algorithm that calculates the value $u(n)$ and stores it in r.
A second event called keep can be also added to simulate some hidden activity before the observation of the event final. Note that the event is *anticipated* and it means that it hides a *ghost* event which will appear later explicitly and effectively. It states that *something like a loop* is acting until the event final.
We have therefore described the problem domain to be solved and we have formulated what we want to calculate. The next step is to inventing a *method of calculation* and this requires a *idea of solution* and the use of refinement.

1.2.1.2. *Refining to compute inductively*

We have defined the specification of the problem of calculating the sum of the elements of a sequence v_0 and we need to find a way to *calculate* the value of the sequence u at term n_0 corresponding to the summation of $v0$. The assignment $r := u(n_0)$ is an expression mixing a variable r and a mathematical value $u(n_0)$. A trivial and inefficient solution is well known: store the values of the sequence u in an array uu

and translate the assignment into the form $r := uu(n)$ where uu verifies the following property $\forall k.k \in dom(t) \Rightarrow uu(k) = u(k)$ and this property constitutes an element of the invariant inv4. The idea is therefore to use the variable uu ($uu \in 0 \,..\, n_0 \pfun \mathbb{Z}$) to control the calculation and its progress. Progression is ensured by the event step2, which decreases the quantity $n - i$ and therefore ensures that the process converges.



The machine *S2* therefore describes a process which progressively fills the table uu and therefore retains all the intermediate results. Proof obligations are fairly easy to prove insofar as we have *prepared* the work of the proof assistant. We will give the details of the statistics in a table at the end of the development. It is quite clear that the variable uu is in fact a witness or a trace of the intermediate values and that this variable can therefore be hidden in this model which will have to be refined. Before hiding this variable, we will set aside the value that we need to keep $uu(i)$.

1.2.1.3. *Focus on the value to be preserved*

The following refinement S3 will lead to the introduction of a new variable cu which will retain the last current value $uu(i)$. We therefore operate a *superposition* (Chandy and Misra 1988) on the machine S2. The idea is therefore that this machine refines or simulates the machine S2 and this also means that properties of the refined machine remain verified by the new machine S3 insofar as the proof obligations are all verified.

MACHINE $S3$ REFINES $S2$ SEES $S0$

VARIABLES $r\ i\ uu\ cu\ n\ v$

INVARIANTS

$@inv1\ cu \in \mathbb{Z}$

$@inv2\ cu = uu(i)$

EVENTS

EVENT $INITIALISATION$

then

$@act1\ r :\in \mathbb{Z}$

$@act2\ i := 0$

$@act3\ uu := \{0 \mapsto 0\}$

$@act4\ cu := 0$

$@act6\ n := n0$

$@act7\ v := v0$

end

EVENT $final$ REFINES $final$

where

$@grd1\ i = n$

then

$@act1\ r := cu$

end

EVENT $step$ REFINES $step$

where

$@grd1\ i < n$

then

$@act1\ i := i + 1$

$@act2\ uu(i + 1) := uu(i) + v(i + 1)$

$@act3\ cu := cu + v(i + 1)$

end

The machine *S3* is very expressive and provides details about the information required to ensure that the machine is suitable for the problem expressed in the machine S1, which is refined by the machine *S2*. It is even clearer that the machine S3 is expensive in terms of variables and the refinement allows us to leave only the variables that are useful for the calculation. In what follows, we will make the machine *more algorithmic* and retain only those variables in the concrete machine that are sufficient for the calculation.

1.2.1.4. *Obtaining an algorithmic machine*

In this final step, we refine the machine *S3* into a machine *S4* and hide the variable uu from the abstract machine *S3*. Thus, the machine *S4* includes variables r, n, v, cu and i and we will also note that it satisfies safety properties called theorems in Event-B in the machine *S4*. These properties are proved from the properties of previous refined machines. We have thus obtained a machine comprising an initialisation and two events:

– The event final4 is observed, when the value of i is n and, in this case, the variable cu contains the value $u(n)$. The invariant guarantees that the value of cu is $u(n)$.

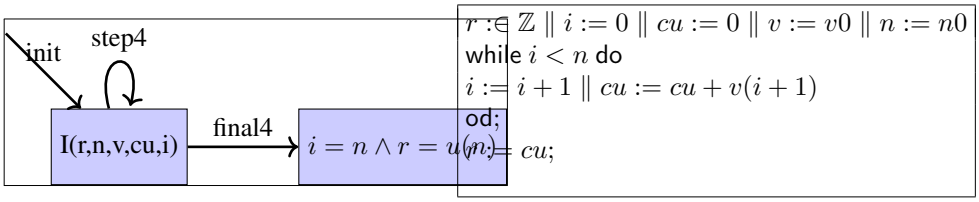
– The event step4 is observed, when the value of i is less than n. This also means that, as long as this value is less than n, the event can be observed and the state transitions generated from these events therefore correspond to *small* steps of an iterative algorithmic structure.

MACHINE $S4$ REFINES $S3$ SEES $S0$
VARIABLES $r\ i\ cu\ n\ v$
INVARIANTS
$theorem\ @inv1\ cu = u(i)$
$theorem\ @inv2\ i \leq n$
EVENTS
EVENT $INITIALISATION$
then
$@act1\ r :\in \mathbb{Z}$
$@act2\ i := 0$
$@act4\ cu := 0$
$@act5\ v := v0$
$@act6\ n := n0$
end

EVENT $final4$ REFINES $final3$
where
$@grd1\ i = n$
then
$@act1\ r := cu$
end

EVENT $step4$ REFINES $step3$
where
$@grd1\ i < n$
then
$@act1\ i := i + 1$
$@act3\ cu := cu + v(i+1)$
end

The Rodin project archive abk-summation corresponds to this development by refinement, taking care to use the calculation method defined by the sequence u. On the left, the following figure describes a view of events observed as labels of transitions from a node containing the loop invariant denoted $I(r, n, v, cu, i)$ and a node containing the postcondition of the contract. The right part contains the algorithm generated by applying rules on events of the machine *S4*.



The components of abk-summation are constructed using the sequence u as a methodological guide, taking care to obtain conditions that can be expressed in an algorithmic language. In our case, the condition $n \notin dom(uu)$ (resp. $n \in dom(uu)$) is refined by the condition $i < n$ (resp.$i = n$). Note that the diagram on the left corresponds to the algorithm on the right. These transformations can be defined more clearly and are implemented in a EB2ALGO (Singh 2024) plugin which produces the above algorithm from the abk-summation archive. Jean-Raymond Abrial (Abrial 2010, Chapter 15) suggests progressive transformation rules to be applied on events like S4 and we will give a more complete treatment of these transformation rules implemented in EB2ALGO (Singh 2024).

1.2.1.5. *Comments on the methodology*

A specific methodology was employed in the selection of the variables. The variable uu is utilised for the storage of the calculated values of the sequence u, with the

convention being to link the sequence u and the variable uu obtained by doubling the name of u. Obviously, we don't want to store all the intermediate values, just the ones used by the induction step. So the variable cu acts as a pointer to the value of uu that is useful in the induction step. uu is a model variable that is no longer necessary to retain for the algorithm. However, it has made the proof work easier, so it should be retained. Hiding uu provides a truly algorithmic view. It is also possible to obtain the termination of this algorithm with minimal effort, thanks to the variant which indicates that the event step4 leads to the convergence of this algorithm. The name final is only imposed by the plugin EB2ALGO (Singh 2024) and the use of Jean-Raymond Abrial (Abrial 2010, Chapter 15).

1.2.2. *Transformations machines Event-B into sequential algorithms*

We take the conclusions of this simple problem and add the extra elements you need to develop iterative sequential algorithms. The plugin EB2ALGO implements the transformations of Jean-Raymond Abrial (Abrial 2010, Chapitre 15). We apply two transformations to the abstract machines obtained at the end of the refinement process, which we called ALGO, and it simplifies the computing process by hiding model variables. We recall the algorithmic language used by Jean-Raymond Abrial (Abrial 2010, Chapter 15)).

Definition 1 *(The Pidgin Programming Language(Abrial 2010))*

Statements of the language are:

– $variable\_list := expression\_list$

– statement; statement

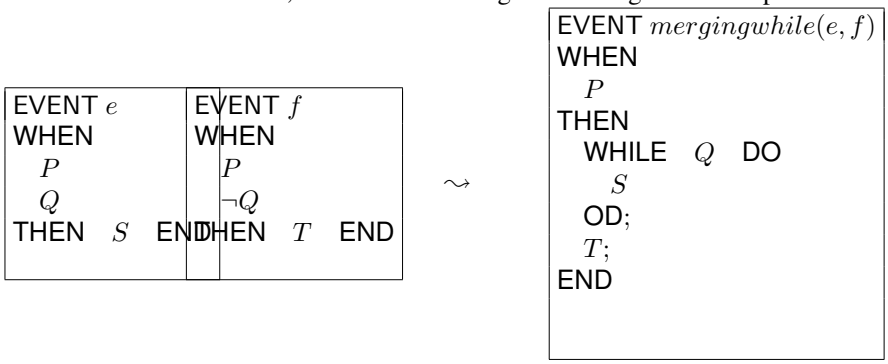
– *if condition* then *statement* else *statement* end

– *if condition* then *statement* elseif ... else *statement* end

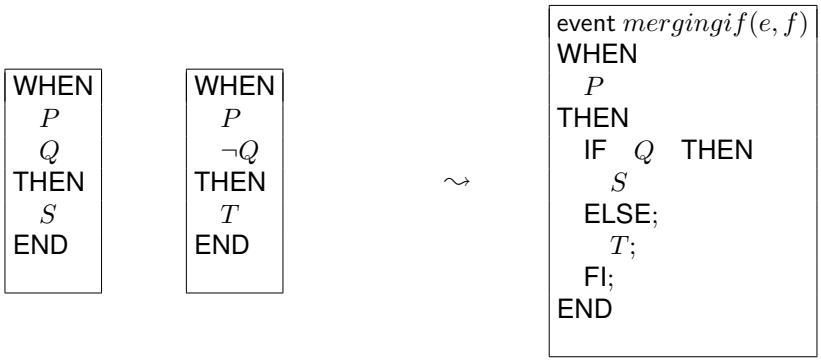
– *while condition* do *statement* end

A program can be *broken down* into a set of events, which are then triggered according to values of the variables. This decomposition leads to the use of a composition of events. The idea is straightforward, but it must adhere to strict conventions to use the plugin EB2ALGO (Singh 2024). We give these conditions which are implemented in the plugin and which must be satisfied during development design. Rules are given without proof of correctness and they are using a mixed language merging Event-B events and programming constructs.

Let us consider two events, which we will merge into an algorithmic expression:



We must specify the conditions of application. The event e appears as new or non-anticipated, therefore convergent at a lower level of refinement than that of f. We can be sure that there is a variant which terminates the loop. We must also assume that P is preserved by the event e. The event mergingwhile(e,f) appears at the same level as the event f. It is possible that P is not present, corresponding to $TRUE$



This transformation should be applied when the two events have been introduced at the same time. The event mergingif(e,f) must appear at the same level as the component. The guard P may not be present.

These two transformations can be used to design a plugin that produces a program in the language Pidgin. The first event is always called final and corresponds to the contract. Then the refinement process guides the design phase and it is also important to express that the new events that are introduced must decrease a variant that ensures the convergence of the process described by the events. The new events are called convergent events and proof obligations are generated to handle the convergence of the events with respect to the variant.

Let us take the example we've already dealt with and apply the transformations.

MACHINE $E-ALGO$
REFINES $D-PREALGO$
SEES $A-C0$
VARIABLES $r\ i\ cu\ n\ v$
INVARIANTS
$theorem\ @inv1\ cu = u(i)$
EVENTS
EVENT $INITIALISATION$
then
$@act1\ r :\in \mathbb{Z}$
$@act2\ i := 0$
$@act4\ cu := 0$
$@act5\ v := v0$
$@act6\ n := n0$
end

EVENT $final$ REFINES $final$
$when$
$@grd1\ i = n$
then
$@act1\ r := cu$
end

EVENT $step000$ REFINES $step000$
$when$
$@grd1\ i < n$
then
$@act1\ i := i + 1$
$@act3\ cu := cu + v(i + 1)$
end

We have given an example of how to apply the merge transformation for iteration (see Fig.:1.2) and we refer the user to the plugin EB2ALGO that implements these transformations. The next section is illustrating the different ways to use the transformations of Event-B models into the Pidgin programming language.

1.3. Examples of development

1.3.1. *Problem 2: Computing the function cube $\lambda x.x^3$ using only additions*

The objective is to calculate the function cube ($\lambda x.x^3$) of a non negative integer using only addition operations. First we define a sequence z corresponding to the cubes of non negative integers. We do not explain how to define the sequence z which is satisfying the correctness property as $\forall n\ in\mathbb{N}.z(n) = n * n * n$. Morever, the inductive definition of z gives a way to compute the term $z(n)$ for any natural number n. Hence, we have a computing method for the function cube using the sequence z.

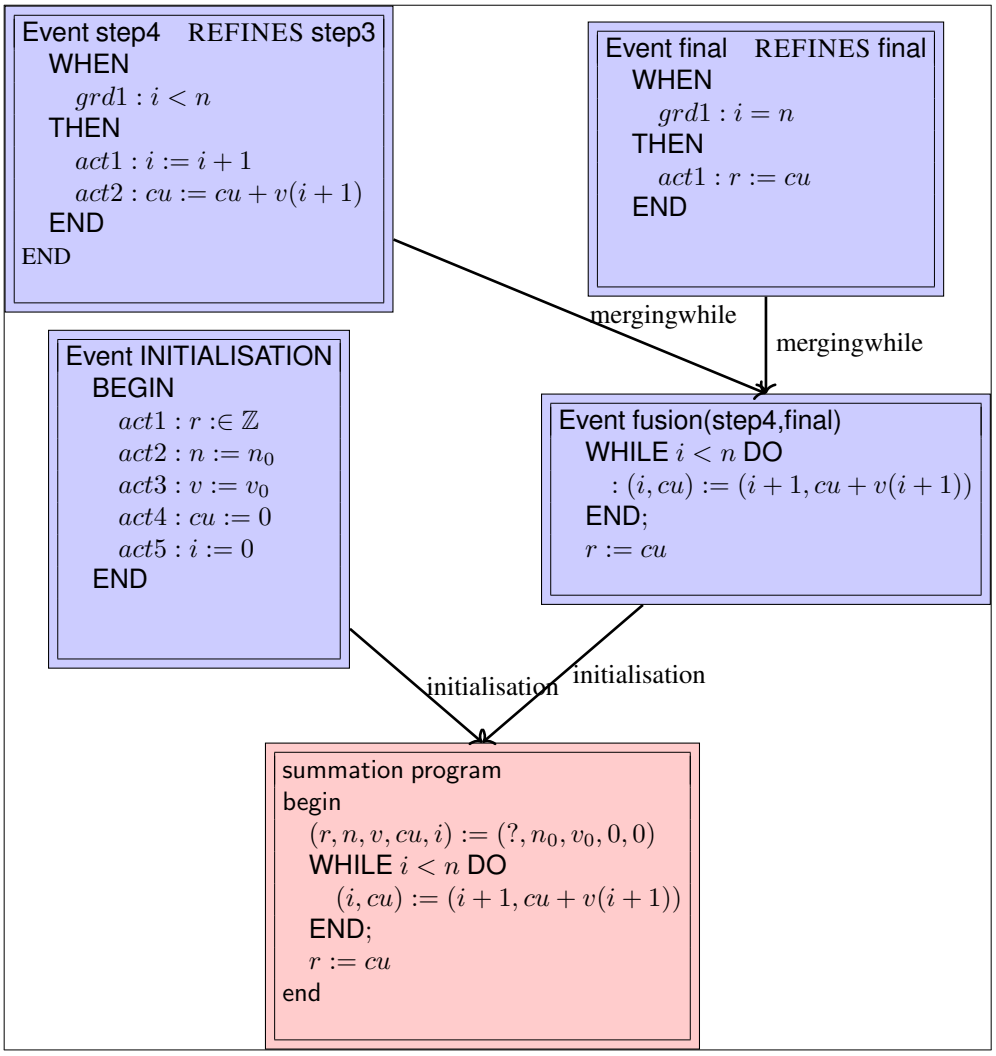
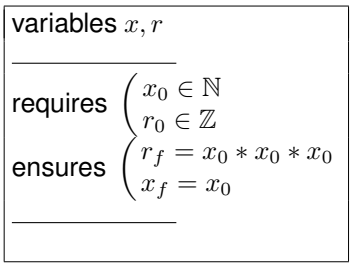


Figure 1.2. *Structure of applied transformations*



Calculating the function cube ($\lambda x.x^3$) using only addition is based on the following sequences:

– $z_0 = 0$ et $\forall n \in \mathbb{N} : z_{n+1} = z_n + v_n + w_n$

– $v_0 = 0$ et $\forall n \in \mathbb{N} : v_{n+1} = v_n + t_n$

– $t_0 = 3$ et $\forall n \in \mathbb{N} : t_{n+1} = t_n + 6$

– $w_0 = 1$ et $\forall n \in \mathbb{N} : w_{n+1} = w_n + 3$

– $u_0 = 0$ et $\forall n \in \mathbb{N} : u_{n+1} = u_n + 1$

The first step is to show that the sequence z defines the sequence of cubes of non negative integers and therefore gives an inductive way of calculating the cube of a non negative integer $x0$ using only addition alone. The domain of the problem to be solved is that of integers $\mathbb{Z}$ and the contract expresses that the value of the result is the cube of the positive integer x_0.

The context P3-0 expresses the sequences defining the values to be calculated to produce a value corresponding to the cube of x_0. The important result is to show the following theorem proved using the Rodin platform.

Propery 1 *(Soundness of the sequence z)*

$$\forall k.k \in \mathbb{N} \Rightarrow z(k) = k * k * k$$

This property guarantees that calculating the terms of the sequence z allows the value to be calculated using auxiliary sequences and only addition operations. These elements are given in the context P3-0. The contract can then be written in the form of a machine. P3-1 contains a single event, final, whose action is `r := z(`x_0`)`.

The method consists in refining the machine P3-1 into a machine P3-2 and introducing an iteration variable i covering the interval $0..x_0$ and variables for each sequence uu, vv, ww, tt, zz whose role is to store values of sequences to calculate $z(x_0)$. A new event is introduced to update the variables uu, vv, ww, tt, zz and i. The invariant expresses the link between the mathematical values of the sequences u, v, w, t, z and the values stored and calculated in the variables uu, vv, ww, tt, zz. The invariant is fairly easy to determine, but we probably need to provide more relationships between the different sequences, so we come back to those sequences which have expressions that only mention i.

Propery 2 *(Properties of sequences u,v,w,t)*

- $\forall k \in \mathbb{N} : v_k = 3 * k * k$
- $\forall k \in \mathbb{N} : w_k = 3 * k + 1$
- $\forall k \in \mathbb{N} : u_k = k$
- $\forall k \in \mathbb{N} : t_k = 3 * k + 3$

The properties are proved in the context P3-0 and will be used in the refinement of the machine P3-2. We introduce variables that point to the elements of the sequences that are sufficient for the computation. The refinement of P3-2 into P3-3 amounts to adding a variable for each sequence cu, cv, cw, ct, cz and these variables verify the following invariant property.

$@inv1\ i \in 0..x0$
$@inv2\ cv = vv(i)$
$@inv3\ cw = ww(i)$
$@inv4\ cz = zz(i)$
$@inv5\ ct = tt(i)$
$@inv6\ cu = uu(i)$

In addition, the events final and step are refined by making guards *computable*. An expression of the form $x0 \in dom(zz)$ or $x0 \notin dom(zz)$ is difficult to translate efficiently into an algorithmic language. Thus, the new guard $i < x0$ implies $x0 \notin dom(zz)$ and the new guard $i = x0$ implies $x0 \in dom(zz)$. The resulting machine is therefore more deterministic and closer to an algorithmic (computable) expression. Proof obligations are discharged automatically

We finish by hiding the variables uu, vv, ww, tt, zz in a refinement machine P3-4 of the machine P3-3. It remains to use the property of sequences that we have proved in the context. The proof effort made in the context of P3-4 pays off when it comes to expressing the invariant properties constituting the loop invariant of the iterative algorithm produced from this machine. from this machine. The variable cu is useless, since $cu = i$. We have translated the machine P3-4 in the form of an ACSL algorithm 1.1 automatically verified by Frama-c. We obtained a cross-check of the algorithm derived from the machine P3-4.

Listing 1.1: ACSL power3.c

```
/*@   requires   0 <= x;
      ensures \result == x*x*x;
      assigns \nothing;
*/
int power3(int x)
{int   r, cz,cv,cu,cw,ct, i;
  cz=0;cv=0;cw=1;ct=3;cu=0;i=0;
        /*@
         @ loop invariant 0 <= i && i  <= x; // inv1
         @ loop invariant ct == 6*i +3; // inv5
         @ loop invariant cv== 3*i*i; // inv2
```

```
        @ loop invariant cw == 3*i+1; // inv3
        @ loop invariant cz == i*i*i; // inv4
        @ loop invariant cu == i; // inv6
        @ loop assigns ct,cz,i,cv,cw,cu; // specific Frama-c
        @ loop variant x-i; // variant in Event-B
*/
  while (i<x)
        {
          cz=cz+cv+cw;
          cv=cv+ct;
          ct=ct+6;
          cw=cw+3;
          cu = cu+1;
          i=i+1;}
  r=cz;return(r);
}
```

The archive bb-power3 contains all the machines used to develop this algorithm. A last comment is related to a useless variable in our algorithm, namely cu. It can be removed from the refinement machine P3-4 by refining one step further. The step is safe by refinement.

1.3.2. *Problem 3: Searching a value in an array*

The problem is to find an occurrence of a value x in an array t of length n. There are no constraints on the array or the search technique.

variables t, n, x, r

---

requires
$$\begin{bmatrix} x_0 \in \mathbb{V} \\ t_0 : 1..n_0 \rightarrow \mathbb{V} \\ r_0 \in \mathbb{Z} \times BOOL \end{bmatrix}$$

ensures
$$\begin{bmatrix} t_f = t_0 \wedge n_f = n_0 \wedge x_f = x_0 \\ x_f \notin ran(t_f) \Leftrightarrow r_f = (0, FALSE) \\ x_f \in ran(t_f) \Leftrightarrow ((prj2(r_f) = TRUE) \wedge (t_f(prj1(r_f)) = x_f)) \end{bmatrix}$$

This problem must be reformulated in the form of a sequence that expresses for a value $i \in 0..n$ whether the array t contains the value x between 1 and i. As soon as the value is found in i, $u(j)$ is equal to $u(i)$ and the value found is i. If the value x is not in the table, then the sequence u is identically equal to a sequence of pairs $(0, FALSE)$. $prj1(x)$ (resp. $prj2(y)$) returns the first (resp. second) value of the pair x.

We will define the sequence u in the context of S-0 and use the same methodology. We simplify now the writing by defining constants as x, t, n since they are endurant entities. The problem is to search, if the value x occurs in the array t and only the entity r is perdurant. The context S-0 contains the definitions of the data t, n, x and the axioms defining the sequence u whose value $u(n)$ is the expected solution.

$ontext\ S-0$
SETS V
CONSTANTS $n\ t\ x\ u$
AXIOMS
$@axm1\ n \in \mathbb{N}$
$@axm2\ t \in 1..n \rightarrow V$
$@axm3\ x \in V$
$@axm4\ u \in 0..n \rightarrow \mathbb{Z} \times BOOL$
$@axm5\ u(0) = 0 \mapsto FALSE$
$@axm6\ \forall i\,.\,i \in 0..n-1 \land t(i+1) = x \land (\forall k.\,k \in 1..i \Rightarrow t(k) \neq x) \Rightarrow u(i+1) = i+1 \mapsto TRUE$
$@axm7\ \forall i\,.\,i \in 0..n-1 \land t(i+1) \neq x \Rightarrow u(i+1) = u(i)$
$@axm8\ \forall i\,.\,i \in 0..n-1 \land t(i+1) = x \land (\exists k.\,k \in 1..i \land t(k) = x) \Rightarrow u(i+1) = u(i)$
$theorem\ @axm9\ \forall i.\,i \in 1..n \land prj2(u(i)) = FALSE \Rightarrow prj1(u(i)) = 0$
$theorem\ @axm10\ \forall i.\,i \in 1..n \land prj2(u(i)) = TRUE \Rightarrow prj1(u(i)) \in 1..n$
$theorem\ @axm11\ \forall i.\,i \in 0..n \land prj2(u(i)) = TRUE \Rightarrow (\exists k.\,k \in 1..i \land t(k) = x)$
$theorem\ @axm12\ \forall i.\,i \in 0..n \land prj2(u(i)) = FALSE \Rightarrow (\forall k.\,k \in 1..i \Rightarrow t(k) \neq x)$
end

The machine S-1 expresses that the desired value is $u(n_0)$ and in fact expresses the contract for this problem. Unlike the previous cases, the machine S-1 is refined into a machine S-2 which introduces the variables i and uu defining the inductive calculation scheme and introducing two events step1 and step2 which simulate a conditional instruction according to the axioms defining u. The process continues with the introduction of the value of the sequence uu useful for the calculation, i.e. cu in the machine S-3. The machine S-4 hides the variable uu and produces a fairly classic algorithm.

$r := (0, FALSE) \parallel k := 0 \parallel cuu := (0, FALSE)$
while $k < n$ do
  if $t(k+1) = x$ then
    if $prj2(cuu) = FALSE$ then
      $k := k+1 \parallel cuu := (k+1, FALSE$
    else
      $k := k+1 \parallel cuu := (k+1, TRUE)$
    fi
  else
    $k := k+1 \parallel cuu := cu$
  fi
od;
$r := cuu;$

$r := (0, FALSE) \parallel k := 0 \parallel cuu := (0, FALSE)$
while $k < n$ do
  if $t(k+1) = x$ then
    if $prj2(cuu) = FALSE$ then
      $cuu := (k+1, FALSE$
    else
      $cuu := (k+1, TRUE)$
    fi
  fi
  $k := k+1;$
od;
$r := cuu;$

The db-searching archive contains all the machines used to develop this algorithm. Note that we apply two times the merging of events to obtain two conditional statements: first step1 and step3 and second mergingif(step1,step3) and step2, finally the iteration rule. It is clear that a rewriting of the current algorithm leads to simplify the body of the iteration.

1.3.3. *Problem 4: Computing a primitive recursive function*

Recursive primitive functions correspond to bounded iterations and a recursive primitive function is constructed from a scheme using two recursive primitive functions to define a third function. Examples of such functions are addition, multiplication, division, primality, ... In a way, it is the general case restricted to bounded iteration.

variables x, n, r

services
$H \in \mathbb{N} \times \mathbb{N} \times \mathbb{N} \to \mathbb{N}$
$G \in \mathbb{N} \to \mathbb{N}$
$F \in \mathbb{N} \times \mathbb{N} \to \mathbb{N}$
$F = PRIMREC(G, H)$

requires $x_0 \in \mathbb{N}$, $n_0 \in \mathbb{N}$
ensures $x_f = x_0 \wedge n_f = n_0$, $r_f = F(x_0, n_0)$

The problem is to calculate the function F defined using two other functions G and H which are calculated by algorithms. F is defined by the following equation: for any natural values x,y, $\begin{cases} F(x,0) = G(x) \\ F(x; y+1) = H(x, y, F(x, y)) \end{cases}$
The function F is defined using G and H, which are themselves defined by the same computability schemes. The function F is itself a sequence which is specialised in v. We obtain the following algorithm.

$r :\in \mathbb{Z} \parallel k := 0 \parallel cv := G(x)$
while $k < x$ do
$k := k + 1 \parallel$
$cv := H(x, k, cv)$
od;
$r := cv$

The eb-primitiverecursive archive contains all the machines used to develop this algorithm.

Jean-Raymond Abrial (Abrial 2010, Chapter 15) gives a list of sequential algorithms built using the two transformations and provides the Rodin[1]archives. It is important to note that Jean-Raymond Abrial's examples begin with a machine with an event final modelling the observation of a procedure call corresponding to the problem solved, but also an event progress whose status *anticipated* means that events will appear later to model one or more future loops. The event progress models the implicit and hidden presence of intermediate computations which must respect the invariant of the abstraction level. The event *conserves* or *preserves* the calculation and its invariant; we sometimes call it keep as opposed to an event which is always present called skip. Finally, it is quite clear that the translation still requires an intervention to produce an executable program as we have shown with the 1.1 algorithm and this allows us to check that the translation is correct. We therefore use Frama-C to verify that the resulting algorithm is partially correct with respect to the contract. The loop invariant is derived from the Event-B development. Jean-Raymond Abrial uses Hoare triples to express the specification of the problem to be solved and we prefer to use contracts that are available in programming languages such as ACSL/Frama-C or Spec#/Boogie. Readers of the section may ask for the soundness of the transformations and it is clear that no proof of correctness has been written. However, the correctness appears to be clear and even obvious and remains to be written.

1. The link `https://web-archive.southampton.ac.uk/deploy-eprints.ecs.soton.ac.uk/122/index.html` gives a list of sequential program developments with a tutorial detailing how to refine and what to transform.

1.4. Design of a Recursive Sequential Algorithm

In this section, we will apply the simple idea of analysing the diagram in figure 1.1 to develop a *recursive* algorithm. We promote the one-shot refinement. In the previous section, we refine as long as necessary, especially as long as we obtained a set of events corresponding to observed computations. We have produced the EB2RC plugin (Cheng *et al*. 2016) , which implements this idea. In the diagram, expressions such as start, reccall. and end mean that control is at start (resp. recall, end), when start (resp. recall end) is true. The expression reccall indicates that control is after the call to the procedure P in question, the name of which appears as an event named $P(x-1, tr) : x \neq 0$. observing this call.

1.4.1. *The "Call as Event" Idea*

The refinement-based design of iterative sequential algorithms uses a sequence of values in a domain $\mathcal{D}$. The computation process is based on the recording of the values of the sequence. In the case of the call-as-event paradigm, the pattern is based on the link between the occurrence of an event and a call of a function (or procedure or method) satisfying the pre-condition and post-condition respectively at the call point and the return point. The context C0 defines the sequence of values and the definition of the sequence is used as a guide for the shape of events. The definitions of sequence are reformulated by a diagram which is simulating the different cases when the procedure under development is called namely $P(x, r)$.

The diagram in Fig. 1.3 is derived from the Event-B machine ALGOREC and is a finite state diagram. We use special names for events in the diagram: P(0,r): $x = 0$ stands for the event observed, when the procedure P(x,r) is called with x=0; P(x-1,tr): $x \neq 0$ models the observation of the recursive call of P; P(x,r): $x \neq 0$ stands for the event observed when the procedure P(x,r) is called with $x \neq 0$. P(0,r): $x = 0$ and P(x,r): $x \neq 0$ are refining the event computing which is observed when the procedure P is called. The red arrows express a case analysis and the blue arrows model events. It includes a liveness proof very close to the proof lattices of Owicki and Lamport (Owicki and Lamport 1982).

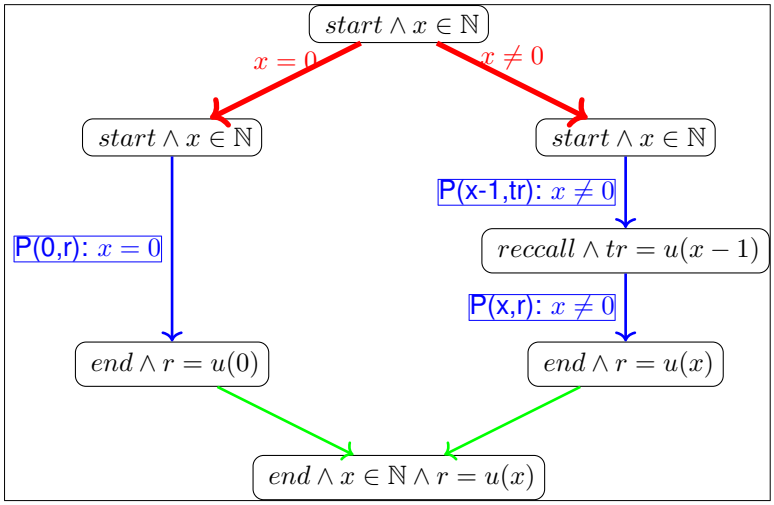


Figure 1.3. *Organisation of the computation in a recursive solution using assertion diagram*

MACHINE $ALGOREC$
REFINES $PREPOST$
SEES $C0$
VARIABLES
r, pc, tr
INVARIANTS
$art : pc \in L$
$inv1 : tr \in D$
$inv2 : pc = callrec \Rightarrow tr = v(x-1)$
$inv3 : pc = end \Rightarrow r = v(x)$

CONTEXT $C0$
SETS D, L
CONSTANTS $v, x, start, end, reccall$
AXIOMS
$ax1 : x \in \mathbb{N}$
$ax2 : v \in \mathbb{N} \rightarrow D$
$ax3 : partition(L, \{start\}, \{end\}, \{reccall\})$
$\ldots$

The refinement machine is organised with respect to the inductive definition using a control variable pc. The variable pc controls the different steps of computations simulated by the events. The invariant is derived directly from the definitions of the intermediate values. Proof obligations are simple to prove. It remains to prove that the values of the sequence v correspond to the required values given in the post-condition.
The machine ALGOREC and the context C0 are given in incomplete form, but the idea is to show that the diagram is linked to the analysis carried out during the computation of the sequence v using a recursive process.

Event P(x,r):x=0
REFINES computing
WHEN
$grd1 : x = 0$
$grd2 : pc = start$
THEN
$act1 : r := d0$
$act2 : pc := end$
END

Event P(x-1,tr):x/=0
WHEN
$grd1 : pc = start$
$grd2 : x \neq 0$
THEN
$act1 : tr := v(x-1)$
$act2 : pc := callrec$
END

Event P(x,r):x/=0
REFINES computing
WHEN
$grd1 : pc = callrec$
THEN
$act1 : r := f(tr)$
$act2 : pc := end$
END

The refinement machine models computations following two cases according to Fig. 1.3. The first case ($x = 0$) is the path on the left part of the diagram and the second case ($x \neq 0$) is the path on the left part.

The first path ($x = 0$) is a three steps path and is labelled by the condition $x = 0$ and the event P(0,r):x=0. The event P(x,r):x=0 is assigning the value $d0$ to r according to the definition of $u(0)$. P(0,r):x=0. refines the event computing in the machine PREPOST. The third step is an implication leading to the postcondition.

The second path ($x \neq 0$) is *four steps* long and the event P(x-1,r):$x \neq 0$ is simulating the recursive call of the same procedure. Finally, the event P(x,r):$x \neq 0$ is refining the event computing. The call as event paradigm is applied, when one considers that one event is defining the specification of the recursive call and the user is giving the name of the call to indicate that the event should be translated into a call. The EB2RC plugin (Cheng *et al.* 2016) generates automatically a C-like program. The machine ALGOREC is simple to check. Proof obligations are simple, because the recursive call is hiding the previous values stored in the variable vv of the iterative paradigm.

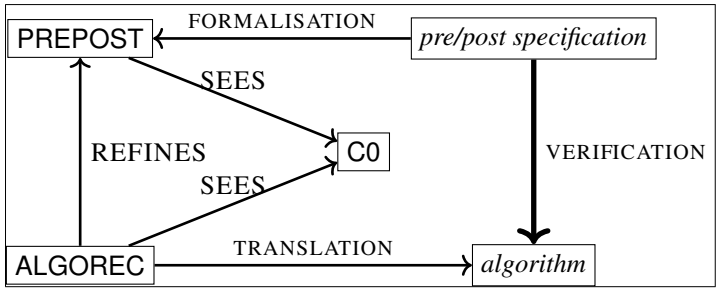
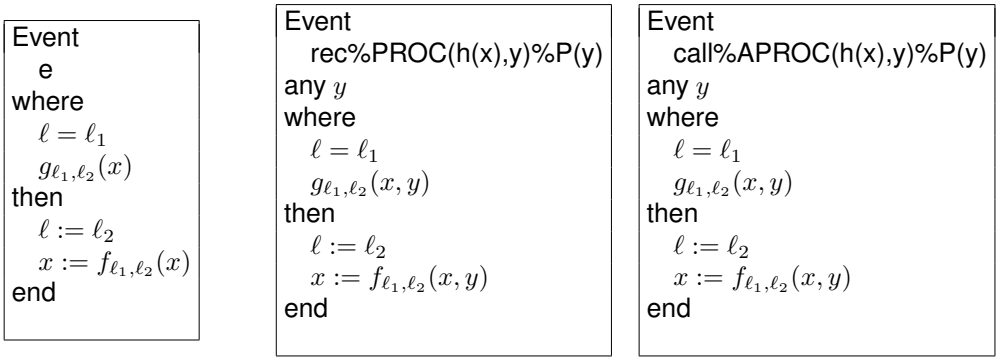


Figure 1.4. *The recursive pattern*

The recursive pattern is linked to a diagram which is helping to structure the solution. We have labelled arrows by guards or by events. The diagram helps to structure

the analysis based on the inductive definitions. Following this pattern, we have developed the ERB2RC plugin based on the identification of three possible events. When a pre/post specification is stated, the program to build can be expressed by a simple event expressing the relationship between input and output and it provides a way to express pre/post specification as events. The first machine contains the event pre/post. The refinement-based process requires an idea for introducing more concrete events. A very simple and powerful way to refine is to introduce a more concrete model which is based on an inductive definition of outputs with respect to the input.

A first consequence is that the concrete model ALGOPC is containing events which are computing the same function but corresponding to a recursive call expressed as events (Event rec%PROC(h(x),y)%P(y)). The event Event rec%PROC(h(x),y)%P(y) is simply simulating the recursive call of the same function and this expression makes the proofs easier. The invariant is defined in a simpler way by analysing the inductive structure and a control variable is introduced for structuring the inductive computation. We have identified three possible events to use in the concrete model:



Each event has a name that follows one of three formats, and the user assigns this name. This convention facilitates the translation of the refinement machine into an algorithm. If the name begins with the character string "rec%...", it corresponds to a recursive call to the current function, and the name ends with the name of the function and the case. Rodin allows any string of characters to be used as an event name. A second case is an event of the form "call%APROC(h(x),y)%P(y)", which indicates that another function or procedure, called APROC, must be used and should reply to the local contract. The name of this function or procedure is indicated by the string of characters following the % symbol. In other cases, the event is translated into an algorithmic expression. The user plays a central role, and the tool checks the consistency of the machine-produced parts. The symbol "%" is used, but another special symbol may be used instead, such as "$".

1.4.2. *Applying the call as event technique*

We will illustrate this method of designing recursive programs using a few relevant examples.

1.4.2.1. *Problem 1: Computing the power 2 ($\lambda x.x^2$)*

Applying the recursive pattern is made easier by the first steps of the iterative pattern. In fact, the context C0 and the machine PREPOST are the starting points of the iterative pattern as well as the recursive pattern. We use the computation of the function x^2 and we obtained the following refinement of PREPOST. Fig. 1.3 is the diagram analysing the way to solve the computation of the value of $u(x)$ following the call-as-event paradigm.

MACHINE $square\ //\ square(n;r)$
REFINES $specquare$ SEES $control0$

VARIABLES $r\ l\ tr$

INVARIANTS
$@inv1\ r \in \mathbb{N}$
$@inv2\ l = end \Rightarrow r = n * n$
$@inv3\ l = call \Rightarrow n \neq 0$
$@inv4\ l = call \Rightarrow tr = (n-1) * (n-1)$
$@inv5\ l \in C$
$@inv6\ tr \in \mathbb{N}$
$@inv7\ l = end \Rightarrow r = n * n$
$@inv8\ l = end \wedge n \neq 0 \Rightarrow tr = (n-1) * (n-1) \wedge r = tr + 2 * (n-1) + 1$
$theorem\ @inv9\ l = call \Rightarrow n * n = tr + 2 * (n-1) + 1$

EVENTS
EVENT $INITIALISATION$
then
$@act1\ r := 0$
$@act2\ l := start$
$@act3\ tr :\in \mathbb{N}$
end

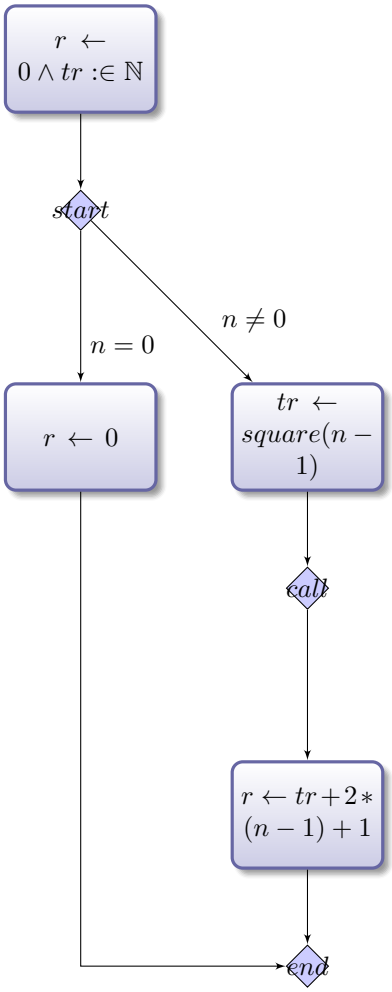
EVENT $square(n;r)@n = 0$
REFINES $square(n;r)$
where
$@grd1\ l = start$
$@grd2\ n = 0$
then
$@act1\ l := end$
$@act2\ r := 0$
end

EVENT $square(n;r)@n/=0$ REFINES $square(n;r)$
  where
  $@grd1\ l = call$
  then
  $@act1\ r := tr + 2*(n-1) + 1$
  $@act2\ l := end$
 end

 EVENT $rec@square(n-1;tr)@n \neq 0$
  where
  $@grd1\ l = start$
  $@grd2\ n \neq 0$
  then
  $@act1\ l := call$
  $@act2\ tr := (n-1)*(n-1)$
 end
end

The variable l is modelling the control in the diagram. We introduce control points corresponding to assertions in the labels of the diagram as $C = \{start, end, callrec\}$. Three events are defined and the invariant is written very easily and proofs are derived automatically. The event rec%square(n-1;tr) is the key event modelling the recursive call. In the current example, we have modified the machine by using directly the fact that $v(n) = n * n$ and normally we had to use the sequence following the recursive pattern and then we had to derive the theorem $v(n) = n * n$.

Proofs are simpler and invariants are easier to extract from the inductive definitions. From the contexts and the machines, by following the rules and by choosing a name for the elements that allow us to produce an algorithm using the EB2RC plugin, we obtain a recursive algorithm that meets the problem specification and we obtain a diagram that shows the different steps of this algorithm.



The diagram is produced with tikz and has annotations defined by this list. The algorithm produced is given below and is very simple to produce from the model

```
procedure square(n; r)
begin
r ← 0
tr :∈ ℕ
if n = 0
      r ← 0
elsif n ≠ 0
      tr ← square(n − 1)
      r ← tr + 2 ∗ (n − 1) + 1
endif
end
```

The invariant states simple and obvious properties related to control points. Theorem 9 is particularly worth examining. It is easy to derive because it corresponds to the recursive call. All the calls and all the details are swept under the carpet, leaving only the last call. This shows the importance and interest of recursion.
The diagram illustrates one functionality of the EB2RC tool.

The invariant states simple and obvious properties related to control points. Theorem 9 is particularly worth examining. It is easy to derive because it corresponds to the recursive call. All the calls and all the details are sweping under the carpet, leaving only the last call. This shows the importance and interest of recursion. The technique is based on the main invariant inv8, which is using the classical identity $\forall n \in \mathbb{N}.n \neq 0 \Rightarrow n^2 = (n-1)^2 + 2 * (n-1) + 1$

MACHINE $square\ //\ square(n; r)$
REFINES $specquare$ SEES $control0$

VARIABLES $r\ l\ tr$

INVARIANTS
$@inv1\ r \in \mathbb{N}$
$@inv2\ l = end \Rightarrow r = n * n$
$@inv3\ l = call \Rightarrow n \neq 0$
$@inv4\ l = call \Rightarrow tr = (n-1) * (n-1)$
$@inv5\ l \in C$
$@inv6\ tr \in \mathbb{N}$
$@inv7\ l = end \Rightarrow r = n * n$
$@inv8\ l = end \wedge n \neq 0$
$\Rightarrow tr = (n-1) * (n-1) \wedge r = tr + 2 * (n-1) + 1$
$theorem\ @inv9\ l = call \Rightarrow n * n = tr + 2 * (n-1) + 1$

1.4.2.2. *Problem 2: Binary search in an array*

We solve the problem of *searching for a value in a table*. The input parameters of the *binsearch* procedure are: a sorted array t; the bounds of the array within which the algorithm should search (lo and hi); and the value for which the algorithm should search (val). Output parameters are $result$ and a boolean flag ok that indicates if $t(result) = val$. The pre and post conditions are presented as follow:

contract $binsearch(t, val, lo, hi, ok, result)$

$$\text{requires} \begin{pmatrix} t \in 0..t.Length \longrightarrow \mathbb{N} \\ \forall k.k \in lo..hi-1 \Rightarrow t(k) \leq t(k+1) \\ val \in \mathbb{N} \\ lo, hi \in 0..t.Length \\ lo \leq hi \end{pmatrix}$$

$$\text{ensures} \begin{pmatrix} ok = TRUE \Rightarrow t(result) = val \\ ok = FALSE \Rightarrow (\forall i.i \in lo..hi \Rightarrow t(i) \neq val \end{pmatrix}$$

EVENT $find$
  any j
  where
  @$grd1$ $j \in lo..hi$
  @$grd2$ $t(j) = key$
  then
  @$act1$ $ok := TRUE$
  @$act2$ $i := j$
end

EVENT $fail$
  where
  @$grd1$ $\forall k \cdot k \in lo..hi \Rightarrow t(k) \neq key$
  then
  @$act1$ $ok := FALSE$
end

The array t is sorted with respect to the ordering over integers and a simple inductive analysis is applied leading to a binary search strategy.

The specification is first expressed by two events corresponding to the two possible cases: either a key exists in the array t containing the value val, or there is no such key. These two events correspond to the two possible resulting *calls* to the procedure $binsearch(t, val, lo, hi; ok, result)$:

– Event find is $binsearch(t, val, lo, hi; ok, result)$ where $ok = TRUE$

– Event fail is $binsearch(t, val, lo, hi; ok, result)$ where $ok = FALSE$

These two events form the machine called *binsearch1* which is refined to obtain *binsearch2* (corresponding to PROCESS of Figure 1.8). In addition to these events, the events of this refined machine contains a new control label, l, which *simulates* how the binary search is achieved.

The refinement of binsearch1 is interesting for showing an invariant derived from the properties of the decomposition following the analysis of the search process. The invariant is explicitly considering the role of the index *middle* and it is a clear statement of the property. We say that the recursion is putting the dust under the carpet and it makes simpler to write and to read the solution.

MACHINE $binsearch2$ REFINES $binsearch1$ SEES $binsearch0$

VARIABLES $i\ ok\ l\ mi$

INVARIANTS

$@inv1\ i \in 1..n$
$@inv2\ l \in LOC$
$@inv3\ dom(t) = 1..n$
$@inv4\ mi \in 1..n$
$@inv5\ l = middle \Rightarrow lo < hi \wedge mi \in lo..hi$
$@inv6\ l = middle \wedge key < t(mi) \Rightarrow (\forall\ k\ k \in mi..hi \Rightarrow t(k) \neq key)$
$@inv7\ l = middle \wedge key > t(mi) \Rightarrow (\forall\ k\ k \in lo..mi \Rightarrow t(k) \neq key)$
$@inv8\ l = end \wedge ok = TRUE \Rightarrow i \in lo..hi \wedge t(i) = key$
$@inv9\ l = end \wedge ok = FALSE \Rightarrow (\forall\ k\ k \in lo..hi \Rightarrow t(k) \neq key)$
$theorem\ @inv10\ lo..hi \subseteq 1..n$
$theorem\ @inv11\ (\exists\ j\ l = middle \wedge j \in mi+1..hi \wedge key > t(mi) \wedge t(j) = key \wedge mi+1$
$\Rightarrow (mi+1 \leq hi)$

The first event is the event INITIALISATION, the two events m1 and m2 corresponding the the case $mo = hi$ and the event split which is corresponding to the case $mo \neq hi$. The events are written the case analysis.

EVENT $INITIALISATION$
then
$@act1\ i :\in 1..n$
$@act2\ ok := FALSE$
$@act3\ l := start$
$@act4\ mi :\in 1..n$
end

EVENT $m1$ REFINES $find$
where
$@grd1\ l = start$
$@grd2\ lo = hi$
$@grd3\ t(lo) = key$
$with$
$@j\ j = lo$
then
$@act1\ l := end$
$@act2\ ok := TRUE$
$@act3\ i := lo$
end

EVENT $m2$ REFINES $fail$
where
$@grd1\ l = start$
$@grd2\ lo = hi$
$@grd3\ t(lo) \neq key$
then
$@act1\ l := end$
$@act2\ ok := FALSE$
end

EVENT $split$
where
$@grd1\ l = start$
$@grd2\ lo < hi$
then
$@act1\ l := middle$
$@act2\ mi := (lo + hi)/2$
end

The event split directs the analysis and divides the exploration space into three parts. Figure 1.4.2.2, the event m3 considers the case where the index mi is the one where the searched value is found. Then the other two events correspond to the segment between mi+1 and hi and are in fact recursive calls to the procedure under construction. These two events refine find and fail respectively. We need to add two more events corresponding to the segment $lo, mi - 1$ segment to complete the model.

A textual representation of the binary search algorithm is constructed by EB2RC. The produced algorithm (as shown in Algorithm **??**) has been compared to the algorithm produced by hand by the authors. The two algorithms are identical up to a slight reformatting.

The proof effort of our refinement approach for the Binary Search case study is illustrated in Table 1.1. The first abstract model is proved automatically and the second concrete model is automatic in 78 % of its proof obligations.

The integrated development framework takes this into consideration. As shown in Fig. 1.8, we suggest to translate every recursive algorithm ALGORITHM into a partially annotated and iterative OPTIMISED ALGORITHM to be verified within the Spec# Programming System. In (Méry and Monahan 2013), we have proposed and

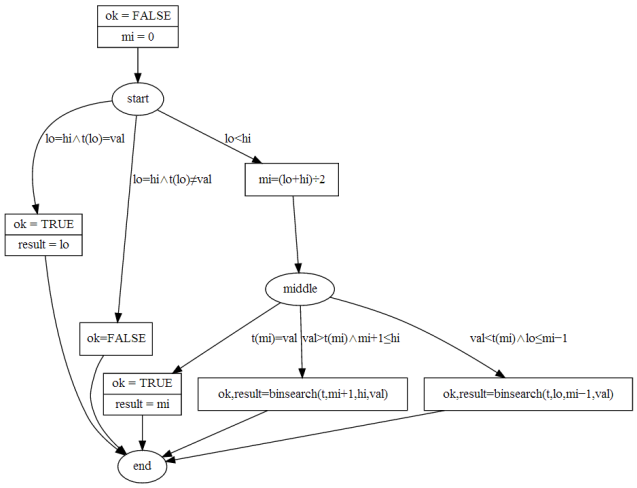


Figure 1.5. *Visualized Representation of the Binary Search Algorithm*

Model	Total	Auto	Manual	Reviewed	Undischarged	% auto
binsearch1	5	5	0	0	0	100 %
binsearch2	71	63	8	0	0	78 %

Table 1.1. *Proof effort of our refinement approach for the binary search case study*

proved a sound translation procedure from ALGORITHM to OPTIMISED ALGORITHM to perform this task. For example, the iterative version of the binary search algorithm in Spec# is shown in Fig. 1.6.

By sending this program to Spec#, Spec# reports the program as verified. No user interaction is required in this verification as all assertions required (preconditions, postconditions and loop invariants) have been generated as part of the refinement and transformation of the initial abstract specification into the final iterative algorithm.

Recursive Algorithm binsearch(t,lo,hi,val;ok,result) generated by EB2RC
begin
$ok := FALSE; mi := 0;$
if $lo = hi \wedge t(lo) = val$ then
  $ok := TRUE;$
  $result := lo;$
elseif $lo = hi \wedge t(lo) \neq val$ then
  $ok := FALSE;$
elseif $lo < hi$ then
  $mi := (lo + hi) \div 2;$
  if $t(mi) = val$ then
    $ok := TRUE;$
    $result := mi;$
  elsif $val > t(mi) \wedge mi + 1 \leq hi$ then
    $ok, result := binsearch(t, mi + 1, hi, val);$
  elsif $val < t(mi) \wedge lo \leq mi - 1$ then
    $ok, result := binsearch(t, lo, mi - 1, val);$
end

1.4.3. *Comments on the call as event idea*

In the example of subsubsection1.4.2.1, we do not use the event like call%APROC(h(x),y)%P(y) but the event is clearly a call for another procedure or function. For instance, when a sorting algorithm is developed, you may need an auxiliary operation for scanning a list of values to get the index of the minimum. It means that we have a way to define a library of models and to use correct-by-construction procedures or functions. In (Cheng *et al*. 2016) , we detail the tool and the way to define a library of *correct-by-construction programs*. The EB2RC plugin is used on this project and we obtain two files: one containing the algorithm and another containing the diagram built from the Event-B model.

1.5. Final comments

We have presented a use of the Event-B language in the derivation of sequential programs or algorithms. The first technique in fact uses the sequences of values leading to a given term constituting the sought-after solution. Thus the calculation of the square or the cube is carried out by defining a sequence one of whose terms is the value of the square or the cube. terms is the value of the square or cube. Insofar as refinement is a very general relationship on a set of very general models too, we must try to set ourselves a refinement discipline. The first technique invites us to play with the model variables and to reduce the variables to those which are exclusively used for

```
class BS {
 int BinarySearch(int[] t, int val, int lo, int hi, bool ok)
  requires 0 <= lo && lo < t.Length && 0 <= hi && hi < t.Length;
  requires lo <= hi && 0 < t.Length;
  requires forall{int i in(0:t.Length),int j in (i:t.Length);t[i]<=t[j]};
  ensures -1 <= result && result < t.Length;
  ensures (0 <= result && result < t.Length)==> t[result] == val;
  ensures result == -1 ==> forall {int i in (lo..hi); t[i] != val};
 { int mi = (lo + hi) / 2;
   while (!(lo == hi && t[lo] == val) || ( lo == hi && t[lo] != val)
                || (lo < hi && (mi == (lo + hi) /2) && t[mi] == val))
    invariant 0 <= lo && lo < t.Length && 0 <= hi && hi < t.Length;
    invariant 0 <= mi && mi < t.Length;
    invariant (val < t[mi]) ==> forall {int i in (mi..hi); t[i] != val};
    invariant (val > t[mi]) ==> forall {int i in (lo..mi); t[i] != val};
   { mi = (lo + hi) / 2;
     if ((mi+1 <= hi) && (val > t[mi])) lo = mi +1;
     else if ((lo <= mi-1) && (val < t[mi])) hi = mi - 1;
   }
   if ((lo == hi) && (t[lo] == val)) {ok = true; return lo;}
   else{
    if ((lo == hi) && (t[lo] != val)) {ok = false; return -1;}
    else if ((lo < hi) && (t[mi] == val)) {ok = true; return mi;}
    else {ok = false; return -1;}
   }
 }
}
```

Figure 1.6. *Binary Search C# program corresponding to the generated iterative procedure.*

the calculation. In this way, we can better understand the notions of model variable or ghost variable used in certain proof tools which look for these variables. In our case, we introduce them and then hide them in the abstract models. They make it easier to prove and state invariants. The second technique is simpler because it relies on inductive definitions that are interpreted in the recursive paradigm. It's about saving refinement and developing at a step of refinement. The resulting program is recursive and the recursiveity should be deleted and it is relatively easy to produce using our event conventions. In this case, we noted that the proofs were relatively simpler to derive. Both techniques rely on experimental plugins but could be combined. Figure 1.8 illustrates an environment for co-ordinating the various techniques for relatively complex sequential systems, but it is necessary to develop a new formal IDE integrating these techniques and close to the diagram in figure. 1.8 . Naturally, the development of concurrent, parallel or distributed algrorithms or programs is a challenge to be taken up, and the chapter entitled *Design of Correct by Construction Distributed Algorithms* will provide some ideas.

EVENT $m3$ REFINES $find$
where
$@grd1\ l = middle$
$@grd3\ t(mi) = key$
$with$
$@j\ j = mi$
then
$@act1\ l\ :=\ end$
$@act2\ ok\ :=\ TRUE$
$@act3\ i\ :=\ mi$
end

EVENT $rec@search(t, val, mi + 1, hi, ok, result)@ok = TRUE$ REFINES $find$
any j
where
$@grd1\ l = middle$
$@grd2\ key\ >\ t(mi)$
$@grd3\ j\ \in\ mi + 1..hi$
$@grd4\ t(j) = key$
$@grd5\ mi + 1\ \leq\ hi$
then
$@act1\ i\ :=\ j$
$@act2\ ok\ :=\ TRUE$
end

EVENT $rec@search(t, val, mi + 1, hi, ok, result)@ok = FALSE$ REFINES $fail$
where
$@grd1\ l = middle$
$@grd2\ key\ >\ t(mi)$
$@grd4\ \forall\ j\ \ j\ \in\ mi + 1..hi\ \Rightarrow\ t(j)\ \neq\ key$
$@grd5\ mi + 1 \leq hi$
then
$@act3\ l\ :=\ end$
$@act2\ ok\ :=\ FALSE$
end

Figure 1.7. *Events for recursive calls*
fig:

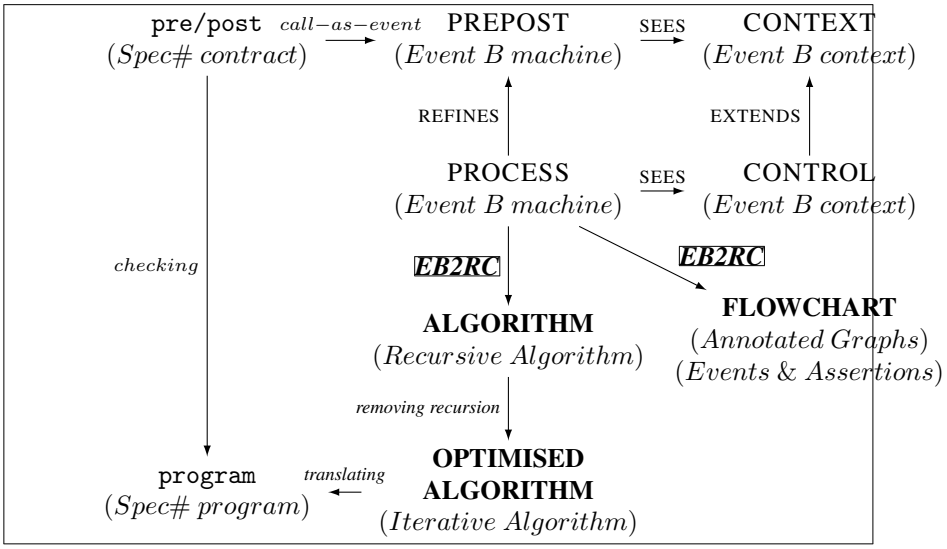


Figure 1.8. *An overview of our integrated development framework to combine program refinement with program verification ((Cheng* et al. *2016))*

1.6. Bibliography

Abrial, J. (2010), *Modeling in Event-B - System and Software Engineering*, Cambridge University Press.
URL: *https://doi.org/10.1017/CBO9781139195881*

Chandy, K. M., Misra, J. (1988), *Parallel Program Design A Foundation*, Addison-Wesley Publishing Company. ISBN 0-201-05866-9.

Cheng, Z., Méry, D., Monahan, R. (2016), On two friends for getting correct programs - automatically translating event B specifications to recursive algorithms in rodin, *in* T. Margaria, B. Steffen, (eds), Leveraging Applications of Formal Methods, Verification and Validation: Foundational Techniques - 7th International Symposium, ISoLA 2016, Imperial, Corfu, Greece, October 10-14, 2016, Proceedings, Part I, vol. 9952 of *Lecture Notes in Computer Science*, pp. 821–838.
URL: *https://doi.org/10.1007/978-3-319-47166-2_57*

Cousot, P. (2021), *Abstract Interpretation*, The MIT Press. ISBN: 9780262044905.

Dijkstra, E. W. (1976), *A Discipline of Programming*, Prentice-Hall.

Floyd, R. W. (1967), Assigning meanings to programs, *in* J. T. Schwartz, (ed.), Proc. Symp. Appl. Math. 19, Mathematical Aspects of Computer Science, American Mathematical Society, pp. 19 – 32.

Hoare, C. A. R. (1969), An axiomatic basis for computer programming, *Communications of the ACM*, 12(10), 576–580.

Kourie, D. G., Watson, B. W. (2012), *The Correctness-by-Construction Approach to Programming*, Springer.
URL: *https://doi.org/10.1007/978-3-642-27919-5*

Méry, D. (2009), Refinement-based guidelines for algorithmic systems, *Int. J. Softw. Informatics*, 3(2-3), 197–239.
URL: *http://www.ijsi.org/ch/reader/view_abstract.aspx?file_no=197&flag=1*

Méry, D., Monahan, R. (2013), Transforming event B models into verified c# implementations, *in* A. Lisitsa, A. P. Nemytykh, (eds), First International Workshop on Verification and Program Transformation, VPT 2013, Saint Petersburg, Russia, July 12-13, 2013, vol. 16 of *EPiC Series in Computing*, EasyChair, pp. 57–73.
URL: *https://doi.org/10.29007/9wm9*

Morgan, C. (1990), *Programming from Specifications*, Prentice Hall International Series in Computer Science, Prentice Hall.

Owicki, S. S., Lamport, L. (1982), Proving liveness properties of concurrent programs, *ACM Trans. Program. Lang. Syst.*, 4(3), 455–495.

Rogers, H. J. (1967), *Theory of Recursive Functions and Effective Computability*, The MIT Press.

Singh, N. K. (2024), EB2ALGO. Plugin Rodin.

Turing, A. (1949), On checking a large routine, *in* Conference on High-Speed Automatic Calculating Machines, University Mathematical Laboratory,Cambridge.